

MISCELLANEOUS CONTROLS

POWERUP RESET: [ESC]P.

QUIT & CLEAR: [ESC]Q.

RESERVE MEMORY: [ESC]R x p s.

SPECIAL CHARACTERS REDEFINE:

[ESC]S a c s p r n f.

	INITIAL VALUE	SPECIAL CHARACTER USAGE
a	ESC	ATTENTION CONTROL (IMMEDIATE)
c	!	NON SPEECH CONTROL (DEFERRED)
s	@	SPEECH CONTROL (DEFERRED)
p	~	PHONEME STRING START
r	CR	RUN EXECUTION BUFFER
n	DC1	XON: RESUME SERIAL SENDING
f	DC3	XOFF: HALT SERIAL SENDING

The last two are SENT TO THE USER when software handshaking is used.

LOAD MEMORY:

!L X DATA. Load exception table

MODE SET: [ESC]M c d s p.

- c = Clock chime (0=off, 1=on, 2=voice)
- d = Delay to auto run (0=off, 1-F sec)
- s = Serial length
- p = Parallel length

USER ROUTINE EXECUTE. !U vars.

WHAT VERSION: [ESC]V.

NOTES: [ESC] denotes the escape code in ASCII (decimal 27) NOT brackets and the letters E,S,C.

The information on this card is believed to be accurate at the date of its publication. However, the Votrax PS System User's manual should be consulted for the current version of this information.

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PHONEME & NOTE TABLES

TABLE 1: PHONEME CHARACTERS

ASC	PHO	ASC	PHO	ASC	PHO	ASC	PHO
@	EH3	P	CH	"	A	0	AW2
A	EH2	Q	SH	!	AY	1	UH2
B	EH1	R	Z	"	Y1	2	UH1
C	PA0	S	AW1	#	UH3	3	UH
D	DT	T	NG	\$	AH	4	O2
E	A2	U	AH1	%	P	5	O1
F	A1	V	001	&	O	6	IU
G	ZH	W	00	'	I	7	U1
H	AH2	X	L	(	U	8	THV
I	I3	Y	K	)	Y	9	TH
J	I2	Z	J	*	T	:	ER
K	I1	[	H	+	R	:	EH
L	M	\	G	.	E	<	E1
M	N	]	F	-	W	=	AW
N	B	^	D	.	AE	>	PA1
O	V	-	S	/	AE1	?	END

ALTERNATE ASCII CHARACTERS WITH THE SAME LOWER 6 BITS AS THE ABOVE MAY BE USED.

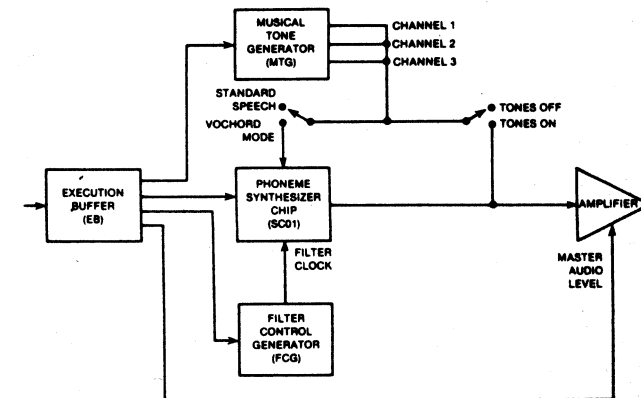
TABLE 2: MUSICAL NOTES 00 = REST

nn	Note	nn	Note	nn	Note	nn	Note
01	C 1	25	C 3	49	C 5	73	C 7
02	C#1	26	C#3	50	C#5	74	C#7
03	D 1	27	D 3	51	D 5	75	D 7
04	D#1	28	D#3	52	D#5	76	D#7
05	E 1	29	E 3	53	E 5	77	E 7
06	F 1	30	F 3	54	F 5	78	F 7
07	F#1	31	F#3	55	F#5	79	F#7
08	G 1	32	G 3	56	G 5	80	G 7
09	G#1	33	G#3	57	G#5	81	G#7
10	A 1	34	A 3	58	A 5	82	A 7
11	A#1	35	A#3	59	A#5	83	A#7
12	B 1	36	B 3	60	B 5	84	B 7
13	C 2	37	C 4	61	C 6	85	C 8
14	C#2	38	C#4	62	C#6	86	C#8
15	D 2	39	D 4	63	D 6	87	D 8
16	D#2	40	D#4	64	D#6	88	D#8
17	E 2	41	E 4	65	E 6	89	E 8
18	F 2	42	F 4	66	F 6	90	F 8
19	F#2	43	F#4	67	F#6	91	F#8
20	G 2	44	G 4	68	G 6	92	G 8
21	G#2	45	G#4	69	G#6	93	G#8
22	A 2	46	A 4	70	A 6	94	A 8
23	A#2	47	A#4	71	A#6	95	A#8
24	B 2	48	B 4	72	B 6	96	B 8

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PERSONAL SPEECH SYSTEM Quick Reference Card



AUDIO CENTER DIAGRAM

Votrax Incorporated
1358 Rankin
Troy, Michigan 48083 USA
1-800-521-1350
OR YOUR LOCAL VOTRAX DEALER

Dealer:

*** PRELIMINARY VERSION ***
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SPEECH & SPEECH CONTROL

TEXT TO BE SPOKEN:

Any string of PRINTABLE ASCII CHARACTERS below that does not begin with a control code will be processed into speech. (! is a control code. Also see REDEFINE.)

ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789 = *
abcdefghijklmnopqrstuvwxyz !.?,,:+\$#%-

PHONEMES STRINGS FOR SPEECH OUTPUT:

A string of phoneme characters preceded by the phoneme control code "~" and terminated by a "?" will be sent to the speech chip directly. SEE TABLE 1.

ex. ~[BX&7? speaks "hello"

AMPLITUDE FADE: @Aar

a = master audio level (0-F)
r = fade rate (0-F)

TEXT CONVERSION MODE: @Cm

m = 0 basic translation
1 exceptions on
2 single numbers

SPEECH RATE: @Rr r = 0-F

VOICE MODE: @Vm

m = 0 standard SC01 (default)
2 VOCHORD (tm) mode (tones off)
4 VOCHORD (tm) mode (tones on)

INFLECTION: @i i = 0-7

REAL TIME CLOCK & ALARMS

TIME SET: [ESC]Thhmmss.
SPEAK TIME: [ESC]T.

ALARM SET:

!A # hhmm TEXT. Set alarm 0-7
!A C. CLEAR ALARMS

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MUSIC & SOUND EFFECTS

MUSICAL TONE CONTROL:

!c nn d. Short form
!c nn d r a g. Complete form
c = channel (1,2,3)
nn = note (00 = rest, 1-96 SEE TABLE 2)
d = duration (0-F tempo units)
r = hold (0-F tempo units)
a = amplitude (0 = envelope, 1-F)
g = glide (0 = off, 1-F)

ENVELOPE SHAPE: !E c a d s r w.

c = channel (1,2,3)
a = Attack rate (0-F)
d = Decay rate (0-F)
s = Sustain level (0-F)
r = Release rate (0-F)
w = attack Wait (0-F Ticks)

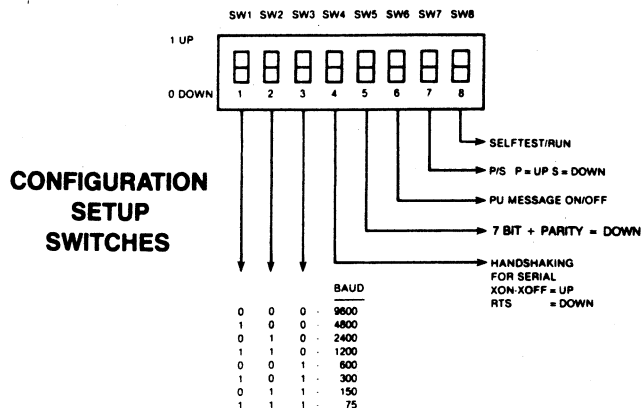
FILTER SETTING: !F bb tt r.

bb = base value (00-FF)
tt = target value (00-FF)
r = rate of movement (0-F)

NOISE: !N c v r.

c = channel (0 = off, 1,2,3)
v = value (A-Z = noise period value)
r = rate of movement (0-F)

TEMPO: !T tt. 1 Tick = 8.192 ms
tt = Tempo unit (00-FF ticks) (ms = milliseconds)



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INPUT / OUTPUT CONTROL

BAUD SELECT: !B b h.

b = 7 75 bps 3 1200 bps
6 150 bps 2 2400 bps
5 300 bps 1 4800 bps
4 600 bps 0 9600 bps
h = handshake 0 = RTS 1 = XON/XOFF

CONNECT INPUT TO: [ESC]C SerIN ParIN.

SerIN, ParIN OPTIONS:

0 = OFF OFFLINE
1 = EB EXECUTION BUFFER
2 = SOUT SERIAL OUTPUT BUFFER
4 = POUT PARALLEL OUTPUT BUFFER

PROMPT:

!P #. Audio prompt 0-9
!P S string. Send string (serial)
!P P string. Send string (parallel)

WAIT FOR:

!W d. Wait d seconds 1-F
!W Sc. Wait for character c (serial)
!W Pc. Wait for character c (parallel)

SPEECH POWER-UP SETTINGS

Amplitude @ AF1
Conversion Mode @ C0
Speech Rate @ R4
Voice Mode @ V0
Inflection @ 1

ERROR CODE DEFINITION

- 1 INVALID CONTROL TYPE
- 2 TERMINATOR MISSING
- 3 INVALID OPTION
- 4 INVALID NUMBER
- 5 MEMORY OVERFLOW
- 6 REQUEST NOT READY
- 7 SERIAL INPUT FAULT

Ref. Card \$1.00

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